Week 1: Module Overview and Python Basics

- **Key Concepts:**
  - **The Environment:** Understanding GitHub Codespaces, the VS Code layout (Explorer, Extensions, Terminal), and the specific structure of a Jupyter Notebook (Markdown vs. Code cells, Kernels).
  - **Data Types:** The distinction between integers (`int`), strings (`str`), floating-point numbers (`float`), lists (`list`), and booleans (`bool`).
  - **Logic & Control Flow:** How computers make decisions (`if...else`) and how they automate repetitive tasks (iteration with `for-loops`).
  - **Zero-based Indexing:** The specific way Python counts positions in sequences (strings and lists).
  - [*Bonus*] **Precision:** The inherent imprecision of floating-point arithmetic (e.g., `0.1 + 0.2 != 0.3`), which is crucial for students to understand early.
- **Packages Introduced:**
  - **No external libraries:** The focus is on the Standard Library modules of Python and the Jupyter Notebook features.
- **Coding Techniques:**
  - **Variable Assignment:** Using `=` to store data and `type()` to inspect it.
  - **Slicing:** Extracting subsets of data from strings and lists (e.g., `[0:4]`).
  - **Formatting:** Using f-strings (formatted string literals) to embed variables into print statements (e.g., `f"Hello, {VariableX}!"`).
  - **Comparison Operations:** Using `==` , `!=` , and `not` to generate Boolean values.
  - **Debugging:** Recognizing `TypeError` when applying incompatible operations (e.g., `len()` on an integer).